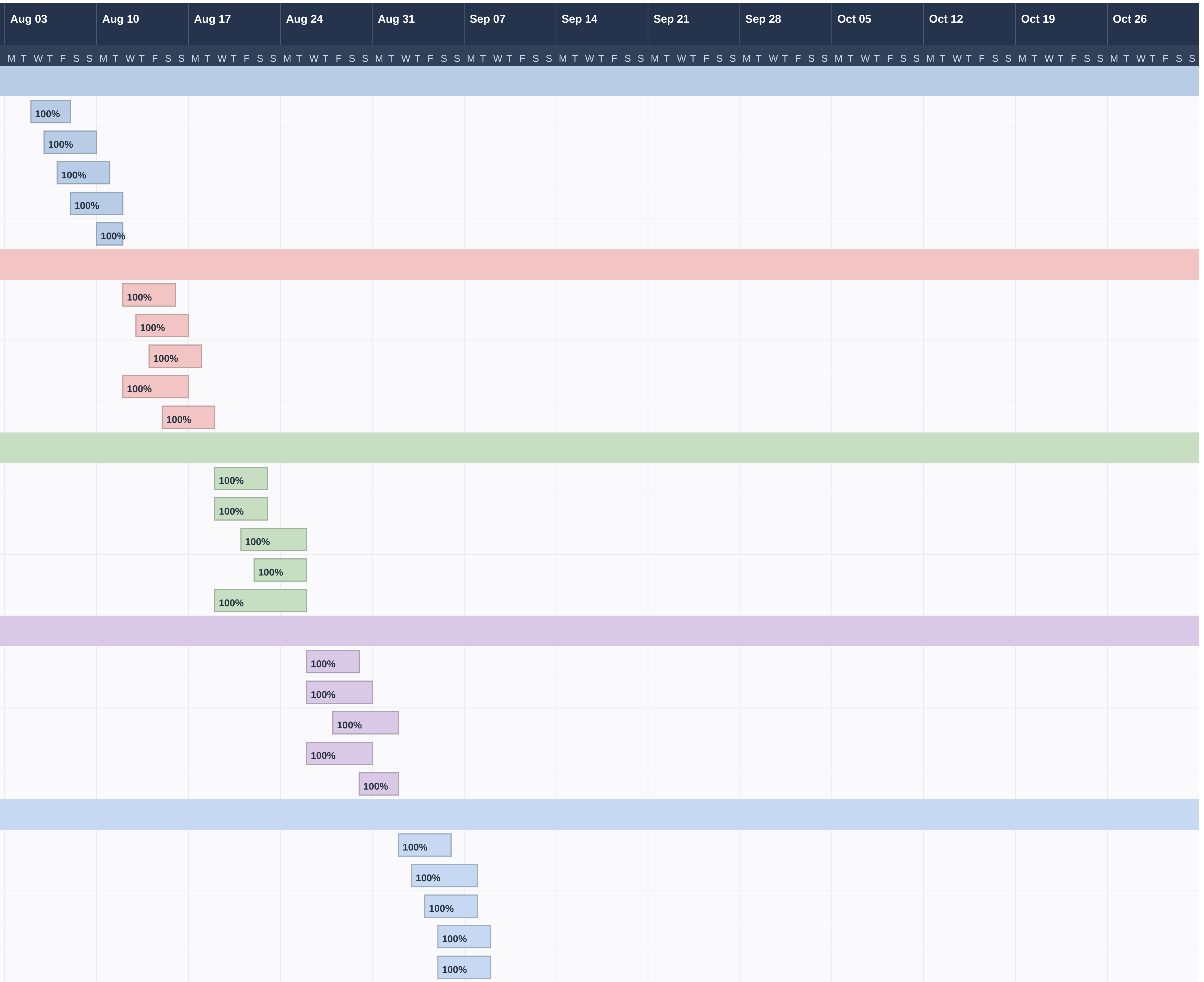
Meridian Spatial Cartographer

UCS503P: Software Engineering (TIET) | Project Schedule & Milestone Tracking

Project Start: Wed, 8/5/2026 Today: Tue, 10/27/2026 Display Week: 1 Page: 1 of 3

Assignee: Rahul Dadwal (Lead), Jyotsna Sachdeva, Bisman Singh Rai

TASK	ASSIGNED TO	PROGRESS	START	END
Phase 1: Team Formation & Problem Identification (Week 1)				
Team Formation & Project Brainstorming	**Team (All Members)**	**100%**	8/5/26	8/7/26
Real-World Problem Analysis & Exploration	**Rahul Dadwal**	**100%**	8/6/26	8/9/26
Meridian Latent Cartographer Concept Study	**Jyotsna Sachdeva**	**100%**	8/7/26	8/10/26
Project Motivation & Problem Domain Exploration	**Bisman Singh Rai**	**100%**	8/8/26	8/11/26
Problem Statement & Direction Finalization	**Team (All Members)**	**100%**	8/10/26	8/11/26
Phase 2: Project Proposal & Requirements Analysis (Week 2)				
Proposal Development & Solution Documentation	**Jyotsna Sachdeva**	**100%**	8/12/26	8/15/26
Functional Analysis & User Interaction Study	**Rahul Dadwal**	**100%**	8/13/26	8/16/26
Data Flow Analysis & Workflow Mapping	**Jyotsna Sachdeva**	**100%**	8/14/26	8/17/26
Presentation Information Gathering & Structure	**Bisman Singh Rai**	**100%**	8/12/26	8/16/26
Presentation Content Drafting (Problem & Goals)	**Bisman Singh Rai**	**100%**	8/15/26	8/18/26
Phase 3: Architectural Modeling & Initial Diagrams (Week 3)				
Initial Use Case Diagram & Actor Analysis	**Rahul Dadwal**	**100%**	8/19/26	8/22/26
Level 0 Context DFD Preparation	**Jyotsna Sachdeva**	**100%**	8/19/26	8/22/26
Level 1 DFD Process & Data Flow Modeling	**Jyotsna Sachdeva**	**100%**	8/21/26	8/25/26
Level 2 DFD Internal Workflow Modeling	**Rahul Dadwal**	**100%**	8/22/26	8/25/26
Project Presentation Slide Deck Complete Draft	**Bisman Singh Rai**	**100%**	8/19/26	8/25/26
Phase 4: Detailed Design & Refinement (Week 4)				
Detailed Use Case Diagram & Interaction Extension	**Rahul Dadwal**	**100%**	8/26/26	8/29/26
Detailed Level 0 & Level 1 DFD Refinement	**Jyotsna Sachdeva**	**100%**	8/26/26	8/30/26
Detailed Level 2 DFD Expansion & Cross-Sync	**Rahul Dadwal**	**100%**	8/28/26	9/1/26
Presentation Updates for New Project Scope	**Bisman Singh Rai**	**100%**	8/26/26	8/30/26
Final Presentation Polish & Submission Readiness	**Bisman Singh Rai**	**100%**	8/30/26	9/1/26
Phase 5: Foundation Prototype - Milestone 1 (Level 0 Implementation) (Week 5)				
Manifest V3 Extension Setup & Local Runtime Architecture	**Rahul Dadwal**	**100%**	9/2/26	9/5/26
Transformers.js & MiniLM WASM Worker Pipeline Setup	**Rahul Dadwal**	**100%**	9/3/26	9/7/26
Data Contracts & Worker IPC Message Protocol	**Jyotsna Sachdeva**	**100%**	9/4/26	9/7/26
Offline Inference Verification & Similarity Testing	**Jyotsna Sachdeva**	**100%**	9/5/26	9/8/26
Demonstration Planning & Overview Review	**Bisman Singh Rai**	**100%**	9/5/26	9/8/26



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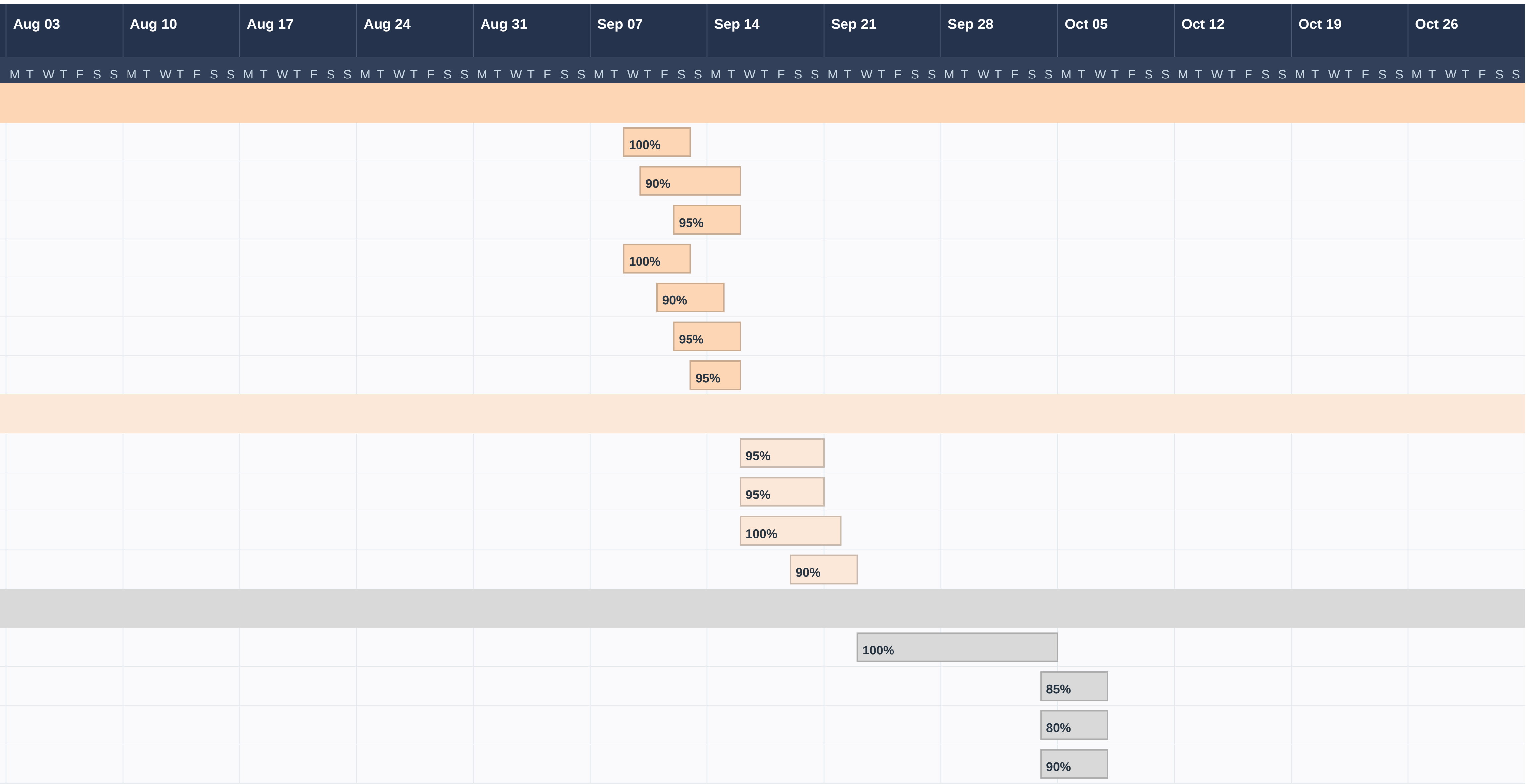
Phase 6: Activity Diagrams, Architecture Implementation & Prototype Report (Week 6)				
Activity Diagram & Swimlane Workflow Modeling	**Rahul Dadwal**	**100%**	9/9/26	9/12/26
Worker Pipeline Optimization & Memory Management	**Rahul Dadwal**	**90%**	9/10/26	9/15/26
Prototype Stage Report (Technical Architecture & Benchmarks)	**Rahul Dadwal**	**95%**	9/12/26	9/15/26
Entity-Relationship (ER) Storage Schema Modeling	**Jyotsna Sachdeva**	**100%**	9/9/26	9/12/26
Offline Latency Profiling & Multi-Sentence Validation	**Jyotsna Sachdeva**	**90%**	9/11/26	9/14/26
Prototype Presentation Technical Architecture Slides	**Jyotsna Sachdeva**	**95%**	9/12/26	9/15/26
Presentation Assembly & Slide Formatting	**Bisman Singh Rai**	**95%**	9/13/26	9/15/26
Phase 7: Mid-Semester Evaluation & Prototype Defense (Week 7)				
Live Prototype Demonstration & WASM Execution Defense	**Rahul Dadwal**	**95%**	9/16/26	9/20/26
Architecture Modeling & DFD/ER Data Flow Defense	**Jyotsna Sachdeva**	**95%**	9/16/26	9/20/26
Evaluation Presentation Delivery & Session Coordination	**Bisman Singh Rai**	**100%**	9/16/26	9/21/26
Evaluator Question Logging & Feedback Synthesis	**Team (All Members)**	**90%**	9/19/26	9/22/26
Phase 8: Mid-Semester Examination Period (Week 8)				
Mid-Semester Theory & Practical Examinations	**Team (All Members)**	**100%**	9/23/26	10/4/26
Technical Architecture Retrospective & Milestone 2 Scope	**Rahul Dadwal**	**85%**	10/4/26	10/7/26
IndexedDB Scalability & Storage Requirements Review	**Jyotsna Sachdeva**	**80%**	10/4/26	10/7/26
Presentation Assets Organization & Archive Management	**Bisman Singh Rai**	**90%**	10/4/26	10/7/26



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Phase 9: Ingestion, Storage & Workflow Architecture Planning (Week 9)				
Readability DOM Parsing & Extension Extraction Research	**Rahul Dadwal**	**85%**	10/7/26	10/10/26
Background Service Worker IPC State Machine Design	**Rahul Dadwal**	**80%**	10/9/26	10/12/26
Sliding-Window Tokenizer (256w/50o) Algorithm Formulation	**Rahul Dadwal**	**75%**	10/10/26	10/13/26
Tiered Client-Side Persistence Architecture (IndexedDB & OPFS)	**Jyotsna Sachdeva**	**85%**	10/7/26	10/11/26
Object Store Schemas & Composite Key Indexing Plan	**Jyotsna Sachdeva**	**80%**	10/9/26	10/12/26
Database Transaction Boundaries & Atomic Save Rollback Rules	**Jyotsna Sachdeva**	**75%**	10/11/26	10/13/26
Team Meeting Minutes & Sprint Schedule Alignment	**Bisman Singh Rai**	**95%**	10/7/26	10/10/26
Cognitive User Journey & Web Research Workflow Study	**Bisman Singh Rai**	**85%**	10/8/26	10/12/26
Spatial Canvas UI Design Pattern Reference Collection	**Bisman Singh Rai**	**90%**	10/10/26	10/13/26
Phase 10: Canvas Layout, Scalability & Retrieval Strategy Planning (Week 10)				
Dynamic 2D Spatial Layout & UMAP Algorithm Research	**Rahul Dadwal**	**80%**	10/14/26	10/17/26
Non-Blocking UMAP Projection Web Worker Architecture	**Rahul Dadwal**	**75%**	10/16/26	10/19/26
HTML5 Canvas / Konva Viewport Rendering Pipeline (60 FPS)	**Rahul Dadwal**	**70%**	10/17/26	10/20/26
In-Memory Cosine Similarity & SQ8 Quantization Algorithm	**Jyotsna Sachdeva**	**80%**	10/14/26	10/18/26
Data Lifecycle Rules & 7-Day Snapshot Cache Pruning Policy	**Jyotsna Sachdeva**	**85%**	10/16/26	10/19/26
Automated Test Strategy & Edge-Case DOM Fixtures Specification	**Jyotsna Sachdeva**	**75%**	10/17/26	10/20/26
Non-Technical Usability Analysis & Tab Capture Friction	**Bisman Singh Rai**	**85%**	10/14/26	10/17/26
Card Density & Floating Toolbar Usability Review	**Bisman Singh Rai**	**80%**	10/16/26	10/19/26
Sprint Task Checklists & Journal Log Synchronization	**Bisman Singh Rai**	**90%**	10/18/26	10/20/26
Phase 11: Class & Sequence Modeling, OOP Implementation, Sequence Flow & UX Redesign (Week 11)				
System Class Diagram Design (Core Worker & Controller Classes)	**Rahul Dadwal**	**90%**	10/21/26	10/24/26
System Sequence Diagram Modeling (End-to-End Pipeline)	**Rahul Dadwal**	**85%**	10/22/26	10/25/26
OOP Class Architecture Implementation (EmbeddingClientManager, WorkerMessageController)	**Rahul Dadwal**	**80%**	10/24/26	10/27/26
Storage Entity Class Diagram Design (DocumentRepository, StorageAdapter)	**Jyotsna Sachdeva**	**90%**	10/21/26	10/24/26
Sequence Diagram Event Orchestration & Runtime Message Flows	**Jyotsna Sachdeva**	**85%**	10/22/26	10/25/26
Asynchronous Sequence Implementation (Promise Correlation & Dispatch)	**Jyotsna Sachdeva**	**75%**	10/24/26	10/27/26
2D Research Canvas Wireframes & Ideation Redesign	**Bisman Singh Rai**	**85%**	10/21/26	10/24/26
Spatial Minimap & Floating Navigation Dock Mockups	**Bisman Singh Rai**	**80%**	10/23/26	10/26/26
Contextual Search Modal Layout & Theme Accent Specifications	**Bisman Singh Rai**	**85%**	10/24/26	10/27/26